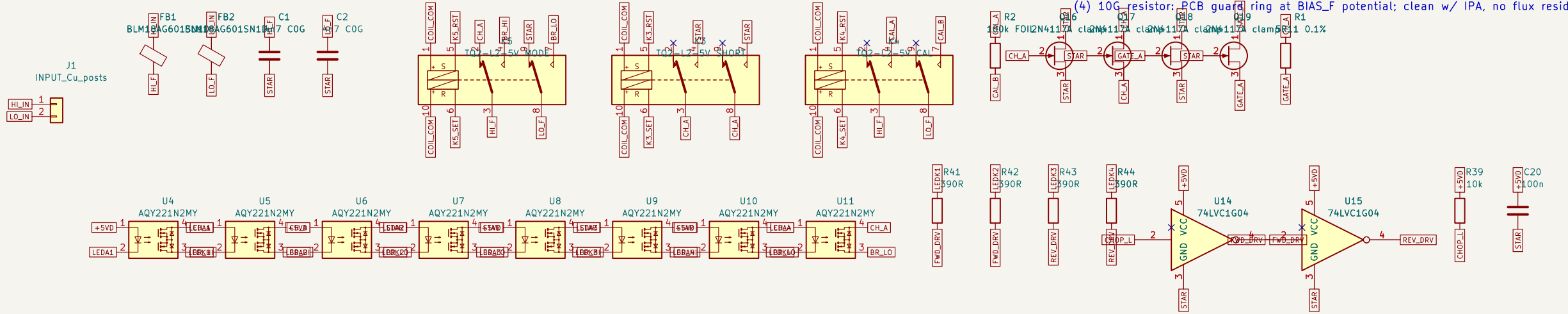


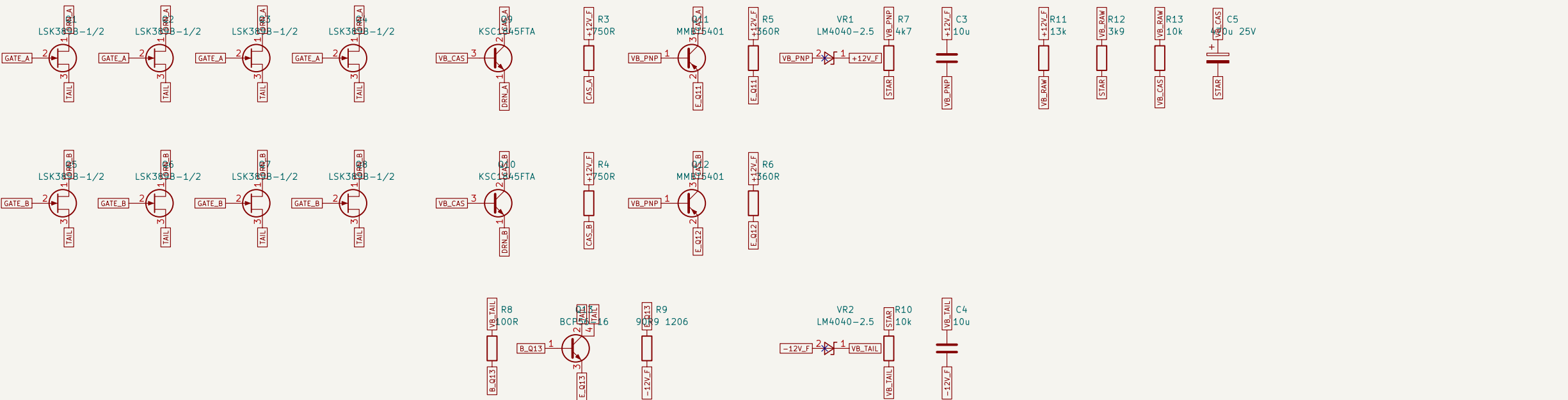
SEC 1 INPUT: isothermal Cu block. K5=mode: direct (high-Z) / SSR bridge (low-Z). SSR chop U4–U11 (2x parallel AQY221N2M per leg) to 100Hz – no contact EMF, no wobble. (1) LSK3898: bin V_{GS} @5mA to <5mV; Cu thermal clamp over all 4 cans. IF3602 (InterFET direct) drops en 0.6–>0.3nV/rHz if sourced.

(3) Signal path HI_IN.GATE_A all–Cu, crimped; no Sn joints inside isothermal block. Guard = STAR, single point.

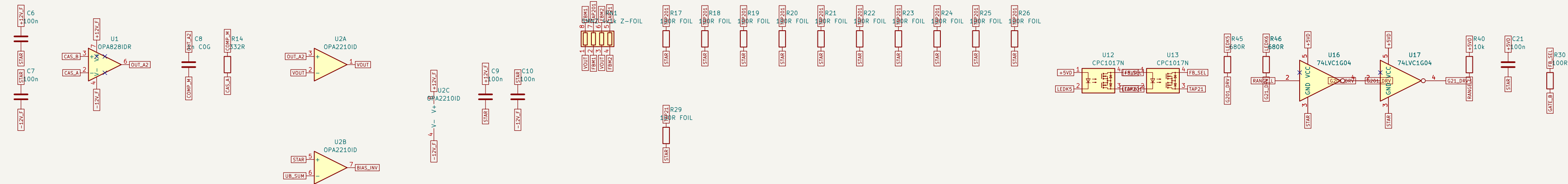
(4) 10G-resistor: PCB guard ring at BIAS_F potential; clean w/ IPA, no flux residue.



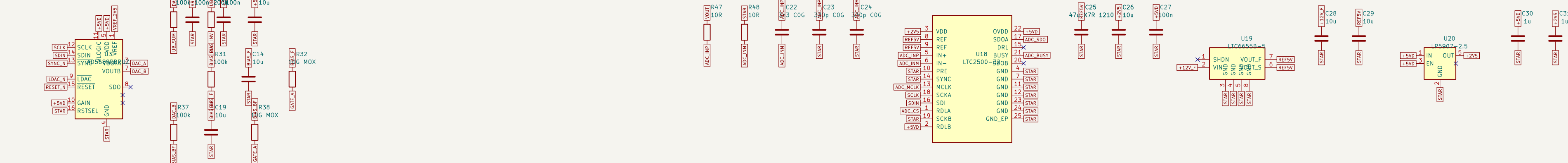
SEC 2 LSK3898 x8 (4 duals): Q1–Q4 gate A, Q5–Q8 gate B. Id=2.5mA each (L_{DSS}>=6mA margin), tail 20mA. Cascode VB=2.77V -> V_{DS}=1.7V. Loads 750R + 5mA bleed.



SEC 3 OPA828 integrator + OPA2210A buffer in loop (designed 1–pole loop: fc 200kHz @G21 / 20kHz @G201, zero 480kHz). RN1 = SMNZ 4x1k: 1+2 = G201 top, 3+4 = G21 top. Bottoms: 10x100R par = 10R (G201), 100R (G21).



SEC 4 BIPOLAR bias servo (leakage flows into gate resistors) DAC chA -> U2B x–2 inverter -> 10G sinks 0..–1nA; DAC chB -> 10G sources 0..+500pA. Filters fc=0.16Hz.



SEC 5 POWER: cap multipliers on both rails. J3 (20–pin) to isolators/drivers on dig board – now incl ADC SPI. SEC 6 (y300): LTC2500–32 + LTC6655 + LP5907 on–board. VOUT never leaves the PCB.

